

BTP Mid-Term Report

Multi-Robot Task Orchestration for Construction Assembly

1. Project Overview

Total Duration: 4 Months

Time Elapsed: 2 Months (August - September)

Time Remaining: 2 Months

Final Goal: To develop a system where multiple robots (e.g., Franka Panda arm, shuttle, mortar dispenser) work collaboratively, orchestrated by an LLM (Large Language Model) brain, to construct a brick wall. Key objectives include precise placement, task orchestration, and seamless multi-robot coordination.

2. Progress Till Now

Based on the current codebase, significant milestones have been achieved in the first two months:

- Simulation Environment: A fully functional PyBullet simulation environment has been established, complete with bricks, pallet stacks, and a pedestal.
- Robot Control & Kinematics: Panda Arm implemented with grasp-error compensation ensuring 1-2mm precision. Shuttle and Mortar Bot developed with collision-safe routing. Robot behaviors are implemented as Python generators allowing concurrent step-by-step execution.
- Task Orchestration (DAG): Developed a task dependency graph and deterministic scheduler that correctly sequences transport, mortar application, and placement tasks.
- Vision Integration: Integrated 3D Computer Vision (OpenCV RGB-D) for overhead camera detection, allowing the system to grasp tilted bricks and perform reachability checks.
- LLM Brain Integration: Initiated development of a 3-Layer LLM Brain (using Gemini models) for strategic planning and tactical supervision.

3. Current Challenges & Problems

- Simplified Physics Interactions: The shuttle currently 'picks up' and 'drops' bricks by teleporting them between the dock and deck. Mortar is simulated as a static slab rather than a fluid extrusion.
- Perception Assumptions: In some headless modes, placement precision assumes perfect perception by reading directly from the simulation state, bypassing real-world vision noise.
- Workspace Constraints: The robotic arm operates in a fixed workspace. Constructing larger walls requires dynamic base placement or a mobile base for the arm.
- Error Recovery Robustness: The closed-loop error recovery orchestrated by the LLM Tactical Supervisor is still under development and needs stress-testing.

4. Future Goals (Next 2 Months)

- Refine LLM Orchestration: Fully transition the mock LLM front-end to a robust LLM brain capable of

BTP Mid-Term Report

Multi-Robot Task Orchestration for Construction Assembly

handling complex, ambiguous natural language commands.

- Enhance Tactical Supervisor: Improve the LLM's ability to monitor the world state and autonomously trigger recovery tools (retry, skip, adjust) without human intervention.
- Improve Physics Realism: Replace teleportation mechanisms with realistic kinematic grasping for the shuttle.
- Comprehensive Benchmarking: Run extensive evaluations to gather metrics on placement accuracy, timeline efficiency, and failure rates under various edge cases.
- Final Project Deliverables: Finalize documentation, prepare final viva defense, and ensure all claims are backed by simulation data.